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Review: Natural transformation.

 $F, G : C \rightarrow D$ Functors $\varphi = F \Rightarrow G$ $\forall A \in \text{Obj}(C) \cdot \varphi_A : F(A) \rightarrow G(A) \in \text{Hom}_D(F(A), G(A))$ s.t. $\forall f \in \text{Mor}(C) \cdot f : A_1 \rightarrow A_2$ $F(A_1) \xrightarrow{\varphi_{A_1}} G(A_1)$ $F(f) \downarrow$ $F(A_2) \xrightarrow{\varphi_{A_2}} G(A_2)$ if every φ_A is an isomorphism(i.e. $\exists f : \text{Hom}_D(G(A), F(A))$ s.t. $f \circ \varphi_A$ $= \text{id}_{F(A)}$, $\varphi_A \circ f = \text{id}_{G(A)}$), φ is

called a natural isomorphism

Two categories C and D are equivalent iff $\exists C \xrightleftharpoons[F]{G} D$ s.t. $G \circ F \simeq \text{id}_C$
 $F \circ G \simeq \text{id}_D$. $\simeq$ means there are some natural isomorphism between these two functorse.g. $C \in \mathcal{C}$, $\tilde{C} := \text{Obj}(\tilde{C}) = \text{Obj}(C) \sqcup \{\tilde{C}_0\}$ $\text{Hom}_{\tilde{C}}(\tilde{C}_1, \tilde{C}_2) = \text{Hom}_C(C_1, C_2)$ $\text{Hom}_{\tilde{C}}(\tilde{C}_0, C) \in \text{Hom}_C(C_0, C)$ $\text{Hom}_{\tilde{C}}(C, \tilde{C}_0) = \text{Hom}_C(C, C_0)$ $\text{Hom}_{\tilde{C}}(\tilde{C}_0, \tilde{C}_0) \in \text{Hom}_C(C_0, C_0)$ Ex. 1: $C \simeq \tilde{C}$ but notice $\text{Obj}(C) \neq \text{Obj}(\tilde{C})$ Monoidal Category is a category with monoidal structure $\otimes$ which is a functor $C \times C \rightarrow C : (A_1, A_2) \mapsto A_1 \otimes A_2$ s.t. $\exists 1 \in \text{Obj}(C)$
 $1 \otimes A \simeq A \simeq A \otimes 1$ And for our interesting we also need it to be associative. ('strict' is not necessary. ref. Mac Lane's theorem) $C \times C \times C \xrightarrow{\text{id}_C \times \otimes} C \times C$ $\otimes \times \text{id}_C \downarrow$
 $C \times C \xrightarrow{\otimes} C$

The most simplest but important monoidal category is Vect with tensor product

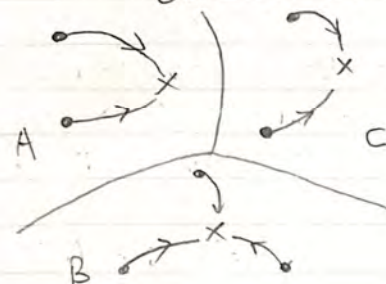
§ MPS/MPO

Matrix product state/operator

We want to model "local" and RG-fixed projection onto g.s. in gapped phase

spin system: $\mathcal{H}^L = \bigotimes_{s=1}^L \mathcal{H}_{\text{local}}$ $H = \sum_s H_{s,s+1}$ 

Phase: Two local systems are in the same phase means they will flow to a same fixed point, i.e. they are in the same Universal class.



1D Ising

$$H = -J \sum_{s=1}^L Z_s Z_{s+1} - h \sum_{s=1}^L X_s \quad (\text{PBC}).$$

Gapped fixed point are typically described by TQFT

Answer: Cobordism Hypothesis

§ 0+1 d

 $\mathcal{H}$: Hilbert space H : Hamiltonian as an adj. operator on $\mathcal{H}$

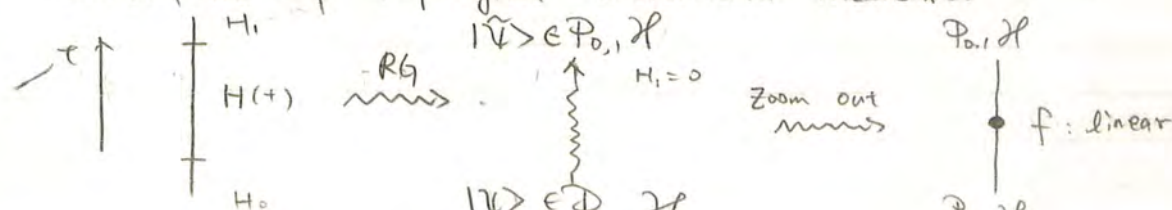
Analog of RG flow: Spectrum cutoff

ground state topological

 $(\mathcal{H}, H) \rightsquigarrow (\mathcal{P}_\Lambda \mathcal{H}, \mathcal{P}_\Lambda H \mathcal{P}_\Lambda^\dagger) \rightsquigarrow \dots \rightsquigarrow (\mathcal{P}_0 \mathcal{H}, \mathcal{P}_0 H \mathcal{P}_0^\dagger = 0)$ E $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff E_0 $\uparrow$ $\sim \sim \sim$ Λ cutoff $\mathcal{P}_0 \mathcal{H}$: Typically finite dimensional Hilbert space
"Topological QM" $Z(\beta) = \text{Tr} e^{-\beta H_0} = \text{Tr} 1 = \dim(\mathcal{P}_0 \mathcal{H}) = \# \text{ g.s.}$

In higher dimension we will characterize gapped fixed points as "locally constructed projection"

Interface of Topological Quantum Mechanics

Finite dimensional vector space: Vect^{fin} $\text{Obj} : \mathcal{P}_0 \mathcal{H} \simeq \text{TQM}$
 $\text{Mor} : \text{Hom}_{\text{Vect}^{\text{fin}}}(\mathcal{P}_{0,0} \mathcal{H}, \mathcal{P}_{0,1} \mathcal{H})$
Interface

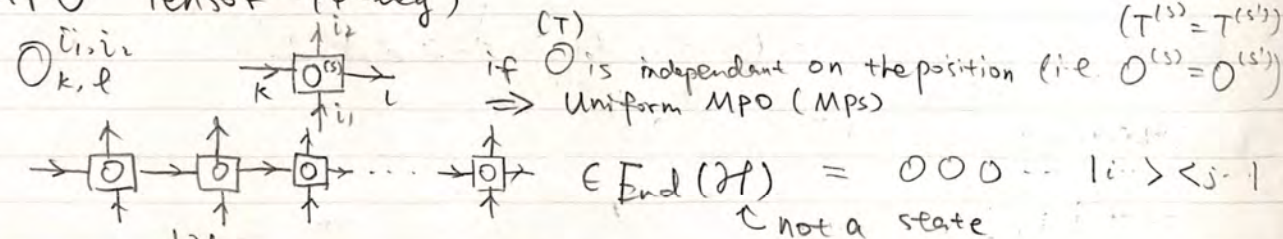
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From the asymptotic behaviour of $S_{A(1)}^{EE}$ we can hope that for MPS

$$D \sim \begin{cases} \text{independent on } L, & \text{gapped} \\ L^\# & \text{gapless} \end{cases}$$

So for general state we need $D \sim d^{1/2}$ but for gapped vacuum state it's possible to simulate it as MPS by using small D .

MPO tensor (4-leg)



T : $\mathcal{H}_B \rightarrow \mathcal{H}_P \otimes \mathcal{H}_B$

O : $\mathcal{H}_B \otimes \mathcal{H}_P \rightarrow \mathcal{H}_P \otimes \mathcal{H}_B$

• SVD (Singular Value Decomposition)

$M: \mathcal{H}_A \rightarrow \mathcal{H}_B$ can be decomposed to $M = V^\dagger \Sigma U$

$U: \mathbb{C}^{\text{rank}(M)} \rightarrow \mathcal{H}_B$ $V: \mathbb{C}^{\text{rank}(M)} \rightarrow \mathcal{H}_A$ $\Sigma: \mathbb{C}^{\text{rank}(M)} \rightarrow \mathbb{C}^{\text{rank}(M)}$

Tensor network is just a network consists of O and T such as

$\mathcal{H}_B \xrightarrow{k} [T] \xrightarrow{i} \mathcal{H}_P \xrightarrow{j} [T] \xrightarrow{l} \mathcal{H}_B$: $\mathcal{H}_B \rightarrow \mathcal{H}_P \otimes \mathcal{H}_P \otimes \mathcal{H}_B \simeq \mathcal{H}_B \otimes \mathcal{H}_B^* \rightarrow \mathcal{H}_P \otimes \mathcal{H}_P$

It can be decomposed to $\mathcal{H}_B \otimes \mathcal{H}_B^* \xrightarrow{V^\dagger} \mathbb{C}^{\text{rank}(M)} \xrightarrow{U} \mathcal{H}_P \otimes \mathcal{H}_P$

$\mathcal{H}_B \xrightarrow{k} [T] \xrightarrow{i} \mathcal{H}_P \xrightarrow{j} [T] \xrightarrow{l} \mathcal{H}_B$ = $\mathcal{H}_B \xrightarrow{k} [V] \xrightarrow{i} \mathcal{H}_P \xrightarrow{j} [\Sigma] \xrightarrow{l} \mathcal{H}_B$ SVD is RG in tensor network

§ RG - fixed MP Projection and Special Frobenius Algebra

goal: systematic construction of MPO's that are

① RG-fixed ② Proj (i.e. $MPO^2 = MPO$)

As examples Special Frobenius Algebra $\rightsquigarrow$ RG - fixed MP projection

① RG-fixed MPO/MPS

① MPS: $\mathcal{H}_B \xrightarrow{k} [T] \xrightarrow{i} \mathcal{H}_P \xrightarrow{j} [T] \xrightarrow{l} \mathcal{H}_B$ $\xrightarrow{\text{SVD}}$ $\mathcal{H}_B \xrightarrow{k} [\tilde{T}] \xrightarrow{i} \mathcal{H}_P \xrightarrow{j} [\tilde{T}] \xrightarrow{l} \mathcal{H}_B$ $\tilde{T} = V^\dagger \Sigma$

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A generic state in $\mathcal{H}^L = \bigotimes \mathcal{H}_{i_l}$ can be spanned as

$$|\psi\rangle = \sum_{i_1, \dots, i_L} \Psi_{i_1, \dots, i_L} |i_1, \dots, i_L\rangle$$

Here we assume $\mathcal{H}_{i_l} = \mathbb{C}^d = \text{Span}_{\mathbb{C}}\{|1\rangle, \dots, |d\rangle\}$, so $\dim \mathcal{H}^L = d^L$

MPS = a nice ansatz of states for vacuum or excited states of local systems

MPS tensor: T_{kl}^i , $i=1, \dots, d$ called physical index $d = \dim \mathcal{H}_{i_l}$
 $k, l=1, \dots, D$ called bond index D is depend on you

$|MPS(T)\rangle := \sum_{k_1, \dots, k_L} T_{k_1 k_2}^{(1) i_1} T_{k_2 k_3}^{(2) i_2} \dots T_{k_L k_1}^{(L) i_L} |i_1, \dots, i_L\rangle$
 (periodic)

$\dim \{T_{kl}^{(s) i}\} = L d D^2 \ll d^L$ (if D is not too big)

If you pick $D \sim d^{1/2} \Rightarrow$ MPS can express any state

Why does it work?

Vacuum has a specific entanglement structure. that is well captured by MPS ansatz

Entanglement Entropy.

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B = \left(\bigotimes_{i \in A} \mathcal{H}_i^{\text{loc}} \right) \otimes \left(\bigotimes_{j \in B} \mathcal{H}_j^{\text{loc}} \right)$

$B = A^c$ given $|\psi\rangle \in \mathcal{H}$ $\xrightarrow{\text{reduced density matrix}}$ $\rho_A = \frac{1}{Z} \text{tr}_B (|\psi\rangle\langle\psi|) \in \text{End}(\mathcal{H}_A)$
 $\uparrow$ pure state $\uparrow$ typically mixed state

$\Rightarrow S_{A|\psi}^{EE} := -\text{Tr}_{\mathcal{H}_A} \rho_{A,|\psi\rangle} \log \rho_{A,|\psi\rangle}$

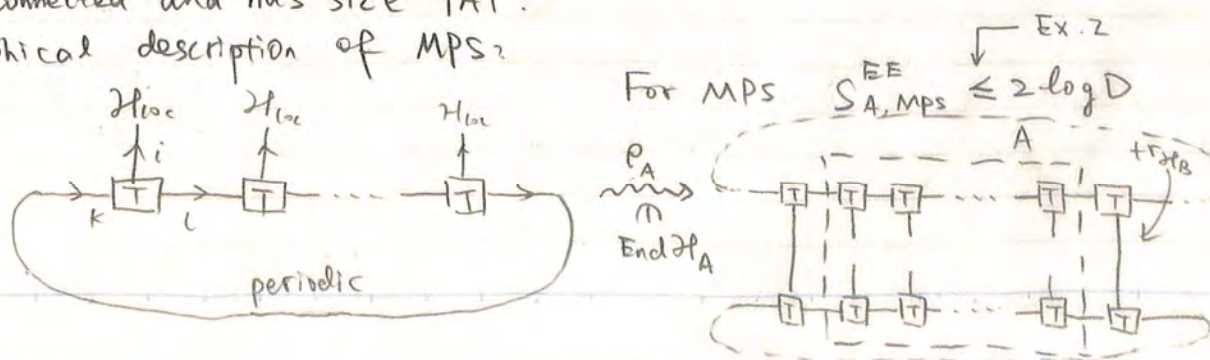
• S^{EE} can measure how close a state is to a product state For product state $S^{EE} = 0$

For a vacuum $|0\rangle$ of local Hamiltonian in $1+d$

$S_{A,|0\rangle}^{EE} \xrightarrow{L \rightarrow \infty} \begin{cases} \text{constant of } |A|, & \text{system is gapped} \\ \frac{c}{3} \log |A| & \text{system is gapless} \end{cases}$

A is connected and has size $|A|$.

Graphical description of MPS:



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As well as the Frob. structure

$$\Delta \circ \mu = \text{id}_A$$

Frob alg. is special if $\mu \circ \Delta = \text{id}_A$

It's not hard to see if $\mathcal{H}_\alpha \approx \mathcal{H}_\beta$ and require $\triangle = \triangle$, we construct a RG-fixed projector. The mathematical structure is just Sp. Frob. Alg. so given a (A, Δ, μ) we can construct a RG-fixed projector O

$$(\mathcal{H}_\alpha = \mathcal{H}_\beta = A) \quad \uparrow = \quad \text{so we can only construct a little special part of MPOs (RG fixed, proj.) by this method.}$$

So actually we don't need ϵ and η to construct O , this is a technical point here we just assume μ and ϵ is demanded for simplicity

Then connect this local O to construct the MPO $\in \text{End}(A^L)$

$$MPO = \dots = \prod_s P_{s, s+1}$$

e.g. of Sp. Frob. Alg. (Ex 3, check) Matrix multiplet

$$A = (\text{Mat}_{n \times n}(\mathbb{C}), \mu, \Delta) \quad \Delta(g^i_j) := \frac{1}{n} \sum_{k=1}^n g^i_k \otimes g^k_j \quad \text{and linear extend it to } \text{Mat}_{n \times n}(\mathbb{C})$$

Thm: Sp. Frob. Alg (with n, ϵ) has an isomorphism.

$$A \cong \bigoplus_{t=1}^{\text{finite}} \text{Mat}_{n_t \times n_t} \quad (\text{Artin-Wedderburn})$$

So Hild is boring. So what are the grand states i.e. $\text{Im } MPO_A$?

It turns out the answer is comodule W over A

A comodule W over A is a vector space with coaction $\Delta_W: W \rightarrow A \otimes W$ and

$$\text{MPS} \rangle \quad \text{Module } \langle \text{MPS} |$$

$$MPO_A |MPS_W\rangle = |MPS_W\rangle$$

RG-fixed MPS: we demand $\tilde{T} = T$ which is possible only when

$$\mathcal{H}_\alpha \approx \mathcal{H}_\beta$$

RG for MPO

RG-fixed MPO: $\tilde{O} = O \Rightarrow \tilde{\mathcal{H}}_\alpha \approx \tilde{\mathcal{H}}_\beta \approx \mathcal{H}_\alpha$

i) projection $MPO^2 = MPO$

$$\text{SVD} \Rightarrow \text{!} \uparrow$$

$\Rightarrow MPO^2$ is also RG-fixed (if MPO is RG-fixed, $\tilde{O} = O$)

2) projection: $MPO^2 = MPO$

$$\text{SVD} \Rightarrow \text{!} \uparrow$$

projecting

$$\tilde{O} = O \quad \tilde{\mathcal{H}}_\alpha \approx \tilde{\mathcal{H}}_\beta \approx \mathcal{H}_\alpha$$

We can obtain RG-fixed MP Projection from Special Frobenius Alg $(A, \mu, \Delta, \eta, \epsilon)$

① $A \in \text{Vect}$ ② $\mu: A \otimes A \rightarrow A$

④ $\eta: \mathbb{C} \rightarrow A$

⑤ $\epsilon: A \rightarrow \mathbb{C}$

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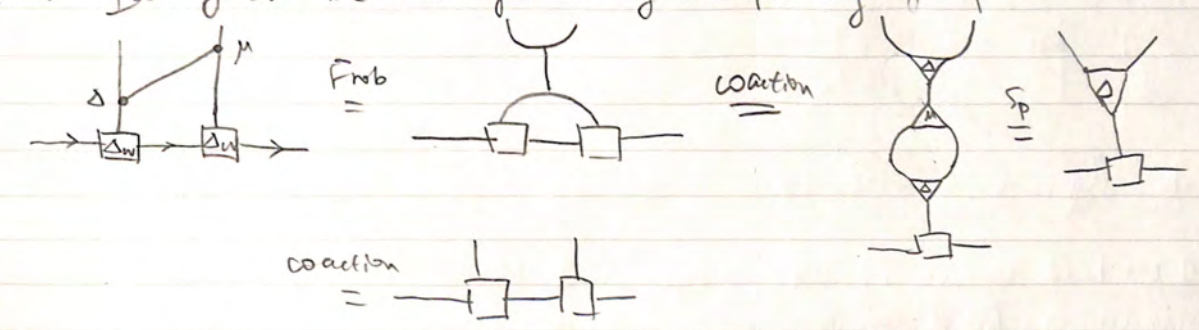
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This equality can be easily seen by the following graph



Ex. $A = \text{Mat}_{n \times n}$, $W = \mathbb{C}^n = \text{span}\{V^{(1)}, \dots, V^{(n)}\}$

$$\Delta_W(V^{(i)}) = \frac{1}{n} \sum_{k=1}^n E^{(ik)} \otimes V^{(k)} \quad E^{ik} := \delta^{ik}$$

$$|\text{MPS}_W\rangle \propto \sum_{k_1, \dots, k_L} E^{k_1 k_2} \otimes E^{k_2 k_3} \otimes \dots \otimes E^{k_L k_1} \in A^L$$

It's called VBS (Valence Bond Solid state)

Ex. 4 : compute S_A^{EE}

Summary: I think it's just tell you how to construct the g.s explicitly by using tensor network

in TOFT